

RTL Design Sherpa

RAPIDS Byte-Granular DMA Engine Byte Characterization Report

September 30, 2026

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Readme

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RAPIDS Byte-Granular DMA: Byte Characterization Report

Version: 0.12
Date: 2026-10-20
Platform: Genesys 2, 256-bit, 8 channels, 100 MHz
Results: `rapids_byte_perf_20261003_122406.json`

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1. Headline

Largest transfers at 8 channels

Path	Point	Efficiency	MB/s measured	% of 3200 MB/s peak
Sink	byte path, 4096 B x 8 ch	1.000	357	11.2 %
Sink	beat path, 4096 beats x 8 ch	1.000	356	11.1 %
Sink	beat path, 4096 beats x 8 ch, word-wide checker build	1.000	3182	99.4 %
Source	byte path, 4096 B x 8 ch	1.000	354	11.1 %
Source	beat path, 4096 beats x 8 ch	1.000	356	11.1 %
Source	beat path, 4096 beats x 8 ch, word-wide checker build	1.000	3199	100.0 %

The byte-wise rates sit at the harness checker ceiling of 3200 / 9 = 355.6 MB/s, not at the DUT's limit: the byte-wise CRC checkers take 9 cycles per 32-byte beat (section 3). Read every MB/s in this report against both the 3200 MB/s peak and that ceiling. Efficiency (payload over beats x lanes) does not depend on the checker.

The word-wide checker build (`BYTE_CRC=0`) takes one beat per cycle and is not checker-bound: its rows show the DUT's own rate against the 3200 MB/s peak. Its data check covers 4 of the 32 bytes of each beat, so it is a rate measurement and the byte-wise build is the integrity measurement.

2. Definitions

Definitions

Term	Meaning
Bytes moved	Total payload bytes over all active channels and descriptors; from the exact byte count of the AXIS bus meter.
Beats moved	Accepted beats on the interface named: AXIS beats on the stream side, memory beats on the AXI side.
Efficiency	payload bytes / (beats x BYTE_LANES), BYTE_LANES = 32. 1.0 means every lane of every beat carried payload.
MB/s	Total bytes / measurement window, windows counted in 100 MHz cycles by the harness meters, not wall clock.
Peak	3200 MB/s = 100 MHz x 32 B, per direction. Shown beside every measured MB/s value.
Engaged utilization	prod / (prod + bp + starv) on one interface; the RAPIDS Beats headline metric, used only for the beat-aligned comparison.

3. Beat-aligned rows against the RAPIDS Beats report

These rows use the beat-scaled path (`pkt_bytes=None`, beats per channel) on the byte DUT, which is how the RAPIDS Beats report measured its matrix: 9-beat bursts, response delay 0, backpressure off, default seed, bare bus meters. The metric is engaged utilization (`prod / (prod + bp + starv)`), the RAPIDS Beats headline metric. "Unchanged" is checked numerically: every cell is compared to the same cell of `genesys_dw256_obs_C.json` and must agree within 0.5 percentage points, the threshold the RAPIDS Beats report itself applies between builds.

3.1 Word-wide checker build (the verdict)

Measured on the `BYTE_CRC=0` bitstream (BUILD.WORD_CRC = 1, results file `rapids_byte_perf_20261003_121944.json`, bitstream sha256 `053fc2c192b869c8`). Its checkers take one beat per cycle like the RAPIDS Beats build. They CRC one 32-bit slice of each beat (slice 0, which holds the beat's LFSR word) and ignore the strobes, so the golden is the RAPIDS Beats golden and the data check covers 4 of the 32 bytes of each beat. That is enough for a utilization measurement of whole-beat rows; full-byte integrity is what the byte-wise build in 3.2 and the rest of this report check. The design under test is the same RTL.

Each meter window opens on its own interface's first handshake (rd, wr, sout), and the sink-ingress window opens on the first ACCEPTED beat; cycles where the stream is offered before that accept are counted by `OBS_SIN_LAUNCH` (CSR 0x158) and are reported with the mechanism table below. The v0.2 report's rows were measured with every window on the shared `obs_dut_busy` open, which charged the descriptor-fetch wait to starvation on the memory-side interfaces; those start-up readings are harness artifacts and were corrected by the window change, not by a DUT change.

Engaged utilization, byte build / RAPIDS Beats reference, beat-aligned rows

Ch	Beats/ch	Sink AXIS-in % (byte / beats)	Sink AXI4-wr %	Source AXI4-rd %	Source AXIS-out %	Max abs delta (pp)
1	1	50.0 / 50.0	50.0 / 14.3	11.1 / 6.7	50.0 / 6.7	43.33
1	4	100.0 / 80.0	80.0 / 40.0	33.3 / 22.2	80.0 / 22.2	57.78
1	16	25.0 / 94.1	94.1 / 72.7	66.7 / 53.3	94.1 / 53.3	69.12
1	64	57.1 / 98.5	98.5 / 91.4	88.9 / 82.1	98.5 / 82.1	41.32
1	256	84.2 / 99.6	99.6 / 97.7	97.0 / 94.8	99.6 / 94.8	15.40
1	1024	95.5 / 99.9	99.9 / 99.4	99.2 / 98.7	99.9 / 98.7	4.38
1	4096	98.8 / 100.0	100.0 / 99.9	99.8 / 99.7	100.0 / 99.7	1.13

Ch	Beats/ch	Sink AXIS-in % (byte / beats)	Sink AXI4-wr %	Source AXI4-rd %	Source AXIS-out %	Max abs delta (pp)
2	1	66.7 / 66.7	50.0 / 22.2	18.2 / 11.8	50.0 / 11.8	38.24
2	4	10.5 / 88.9	88.9 / 57.1	50.0 / 36.4	88.9 / 36.4	78.36
2	16	32.0 / 97.0	97.0 / 84.2	80.0 / 69.6	97.0 / 69.6	64.97
2	64	65.3 / 99.2	99.2 / 95.5	94.1 / 90.1	99.2 / 90.1	33.92
2	256	88.3 / 99.8	99.8 / 98.8	98.5 / 97.3	99.8 / 97.3	11.53
2	1024	96.8 / 100.0	100.0 / 99.7	99.6 / 99.3	100.0 / 99.3	3.16
2	4096	99.2 / 100.0	100.0 / 99.9	99.9 / 99.8	100.0 / 99.8	0.81
4	1	100.0 / 80.0	50.0 / 30.8	30.8 / 19.0	66.7 / 19.0	47.62
4	4	12.9 / 94.1	94.1 / 72.7	66.7 / 53.3	94.1 / 53.3	81.21
4	16	37.2 / 98.5	98.5 / 91.4	88.9 / 82.1	98.5 / 82.1	61.25
4	64	70.3 / 99.6	99.6 / 97.3	97.0 / 94.8	99.6 / 94.8	29.28
4	256	90.5 / 99.7	99.9 / 99.4	99.2 / 98.7	99.9 / 98.7	9.25
4	1024	97.4 / 99.9	100.0 / 99.9	99.8 / 99.7	100.0 / 99.7	2.50
4	4096	99.3 / 100.0	100.0 / 100.0	100.0 / 99.9	100.0 / 99.9	0.64
8	1	4.1 / 88.9	50.0 / 38.1	47.1 / 27.6	80.0 / 27.6	84.83
8	4	14.5 / 97.0	97.0 / 84.2	80.0 / 69.6	97.0 / 69.6	82.42
8	16	40.5 / 99.2	99.2 / 95.5	94.1 / 90.1	99.2 / 90.1	58.72
8	64	73.1 / 99.8	99.8 / 98.5	98.5 / 97.3	99.8 / 97.3	26.66
8	256	91.6 / 96.1	100.0 / 99.7	99.6 / 99.3	100.0 / 99.3	4.51
8	1024	97.8 / 99.0	100.0 / 99.9	99.9 / 99.8	100.0 / 99.8	1.24
8	4096	99.4 / 99.7	100.0 / 100.0	100.0 / 100.0	100.0 / 100.0	0.32

Verdict: CHANGED. 112 of 112 beat-aligned cells were compared; 85 differ by more than 0.5 pp; the largest difference is -84.83 pp (ch 8, 1 beats, sink sin). Cells over the threshold: ch1 b1 sink wr +35.71 pp; ch1 b1 source rd +4.44 pp; ch1 b1 source sout +43.33 pp; ch1 b4 sink sin +20.00 pp; ch1 b4 sink wr +40.00 pp; ch1 b4 source rd +11.11 pp; ch1 b4 source sout +57.78 pp; ch1 b16 sink sin -69.12 pp; ch1 b16 sink wr +21.39 pp; ch1 b16 source rd +13.33 pp; ch1 b16 source sout +40.78 pp; ch1 b64 sink sin -41.32 pp; ….

Where the differences come from. Every difference is a fixed number of cycles per run, not a change of rate. The cycle counts below are the evidence; the same counts at 16 beats per channel are identical to these. Only the 1-beat rows differ: the write and source starvation there equals the channel count (channel count + 7 on the read side), a per-channel start-up term that later beats overlap, and the sink backpressure is 0 at 1 to 4 channels.

Cycle counts at 4096 beats per channel (fixed start-up terms, not rates)

Ch	Sink AXIS-in bp, byte	beats	Sink AXI4-wr starv, byte	beats	Source AXI4-rd starv, byte	beats
1	47	0	1	6	8	14
2	67	0	1	6	8	14
4	107	2	1	6	8	14
8	187	82	1	6	8	14

- **Sink AXIS-in.** The byte ingress accepts a channel's stream only once that channel has a packet record, because the destination offset and the expected length come from the descriptor (sink ingress chapter of the MAS). While one channel occupies the stream the others are offered data with `tready` low, and the meter counts those cycles as backpressure: exactly 27 + 20 x channels, the per-channel packet-record wait of rapids TASK-021, independent of the beats per channel. RAPIDS Beats has no such term because it buffers the stream before the descriptor arrives. The `sin` window opens at the first ACCEPTED beat. The wait this window choice could hide is measured and empty: `OBS_SIN_LAUNCH` (CSR 0x158) counts cycles where the stream is offered before the first accept, and it read 0 on all 28 points of this run (and on all 28 rows of the word-wide sim campaign), so opening at the first offered beat and at the first accepted beat are the same measurement on this DUT.
- **Sink AXI4-wr.** The write window opens at the first write handshake, so the wait above is not in it. Start-up starvation is 1 cycle at 16 beats and up, the same for every channel count (at 1 beat it is the channel count). Under the previous shared window the same cells read 9 to 17 cycles: the descriptor-

fetch wait was being charged to write starvation, and the per-interface windows removed that, not the DUT.

- **Source.** The read and stream-out windows open on their own first handshakes: 8 starvation cycles on `rd` and 1 on `sout` at 16 beats and up, every channel count (channel count + 7 and channel count at 1 beat). These cells read 15 to 22 cycles under the shared window, for the same reason as the write side. The remaining small POSITIVE deltas against the beats reference at short transfers are the same class of fixed start-up term on the beats harness's side of the comparison.
- **Amortisation.** Because each term is fixed, the byte and beats readings converge as the transfer grows: at 4096 beats per channel every cell is within 1.13 percentage points, and the 8-channel row within 0.32. The 1-to-64-beat rows, where a fixed term is a large share of the window, are the ones that move most.
- **Not explained.** With 1 beat per channel and 1, 2 or 4 channels the byte build shows no sink AXIS-in backpressure, while the same channels at 16 beats and up do, and the 8-channel 1-beat row reads 190 cycles against the 187 of the formula. Not isolated.

3.2 Byte-wise checker build (the standard bitstream)

Engaged utilization, byte build / RAPIDS Beats reference, beat-aligned rows

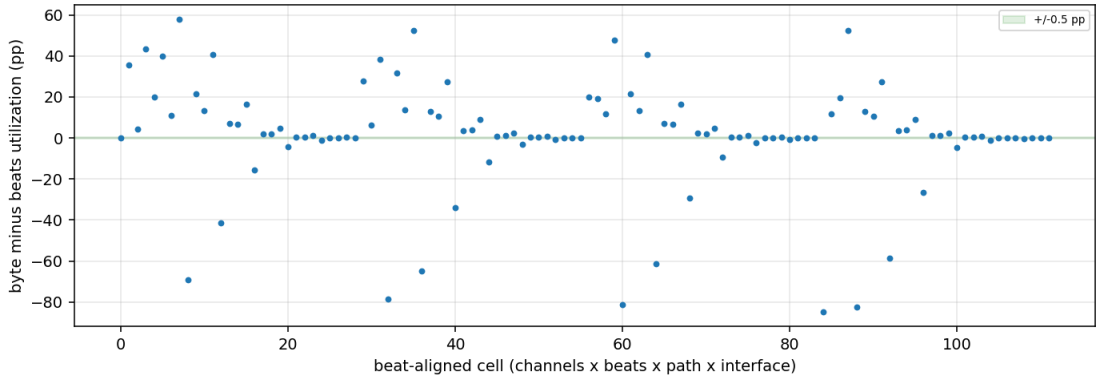
Ch	Beats/ch	Sink AXIS-in % (byte / beats)	Sink AXI4-wr %	Source AXI4-rd %	Source AXIS-out %	Max abs delta (pp)
1	1	50.0 / 50.0	50.0 / 14.3	11.1 / 6.7	100.0 / 6.7	93.33
1	4	100.0 / 80.0	80.0 / 40.0	11.1 / 22.2	14.3 / 22.2	40.00
1	16	25.0 / 94.1	15.7 / 72.7	11.1 / 53.3	11.8 / 53.3	69.12
1	64	57.1 / 98.5	12.0 / 91.4	21.2 / 82.1	11.2 / 82.1	79.44
1	256	22.6 / 99.6	11.3 / 97.7	18.7 / 94.8	11.1 / 94.8	86.39
1	1024	12.7 / 99.9	11.2 / 99.4	12.4 / 98.7	11.1 / 98.7	88.26
1	4096	11.5 / 100.0	11.1 / 99.9	11.4 / 99.7	11.1 / 99.7	88.73
2	1	66.7 / 66.7	50.0 / 22.2	11.1 / 11.8	22.2 / 11.8	27.78
2	4	10.5 / 88.9	26.7 / 57.1	11.1 / 36.4	12.5 / 36.4	78.36

Ch	Beats/ch	Sink AXIS-in % (byte / beats)	Sink AXI4-wr %	Source AXI4-rd %	Source AXIS-out %	Max abs delta (pp)
2	16	32.0 / 97.0	13.0 / 84.2	12.0 / 69.6	11.2 / 69.6	71.20
2	64	65.3 / 99.2	11.5 / 95.5	34.2 / 90.1	11.2 / 90.1	83.99
2	256	14.8 / 99.8	11.2 / 98.8	19.9 / 97.3	11.1 / 97.3	87.63
2	1024	11.8 / 100.0	11.1 / 99.7	12.5 / 99.3	11.1 / 99.3	88.57
2	4096	11.3 / 100.0	11.1 / 99.9	11.4 / 99.8	11.1 / 99.8	88.81
4	1	100.0 / 80.0	50.0 / 30.8	11.1 / 19.0	14.8 / 19.0	20.00
4	4	12.9 / 94.1	15.7 / 72.7	11.1 / 53.3	11.8 / 53.3	81.21
4	16	37.2 / 98.5	12.0 / 91.4	21.3 / 82.1	11.3 / 82.1	79.44
4	64	70.3 / 99.6	11.3 / 97.3	50.2 / 94.8	11.1 / 94.8	86.04
4	256	12.6 / 99.7	11.2 / 99.4	20.8 / 98.7	11.1 / 98.7	88.26
4	1024	11.5 / 99.9	11.1 / 99.9	12.6 / 99.7	11.1 / 99.7	88.73
4	4096	11.2 / 100.0	11.1 / 100.0	11.4 / 99.9	11.1 / 99.9	88.85
8	1	4.1 / 88.9	19.0 / 38.1	11.1 / 27.6	12.7 / 27.6	84.83
8	4	14.5 / 97.0	13.0 / 84.2	11.6 / 69.6	11.1 / 69.6	82.42
8	16	40.5 / 99.2	11.5 / 95.5	35.2 / 90.1	11.2 / 90.1	83.99
8	64	73.1 / 99.8	11.2 / 98.5	66.8 / 97.3	11.2 / 97.3	87.26
8	256	11.8 / 96.1	11.1 / 99.7	21.5 / 99.3	11.1 / 99.3	88.57
8	1024	11.3 / 99.0	11.1 / 99.9	12.6 / 99.8	11.1 / 99.8	88.81
8	4096	11.2 / 99.7	11.1 / 100.0	11.5 / 100.0	11.1 / 100.0	88.87

Result on this build: CHANGED. 112 of 112 beat-aligned cells were compared; 110 differ by more than 0.5 pp; the largest difference is +93.33 pp (ch 1, 1 beats, source sout). Cells over the threshold: ch1 b1 sink wr +35.71 pp; ch1 b1 source rd +4.44 pp; ch1 b1 source sout +93.33 pp; ch1 b4 sink sin +20.00 pp; ch1 b4 sink wr +40.00 pp; ch1 b4 source rd -11.11 pp; ch1 b4 source sout -7.94 pp; ch1 b16 sink sin -69.12 pp; ch1 b16 sink wr -57.04 pp; ch1 b16 source rd -42.22 pp; ch1 b16 source sout -41.57 pp; ch1 b64 sink sin -41.32 pp; ….

What the byte-checker rows can and cannot show. The byte build's on-chip checkers (`BYTE_CRC=1` on `axi4_slave_wr_crc_check` and `axis4_slave_pattern_check`) fold the strobed bytes of each beat into the CRC four bytes per cycle and hold ready low while they do, so a 32-byte beat occupies 9 cycles. That caps the harness at 3200 / 9 = 355.6 MB/s (11.1 % of the 3200 MB/s peak), which is where the large-transfer rows sit. The RAPIDS Beats reference used the word-wide checkers and is not checker-bound. The differences in this table are therefore the checkers backpressuring the DUT, not evidence about the DUT's own throughput, and this table is **not comparable** to the RAPIDS Beats report. The "utilization unchanged" criterion is settled in section 3.1 on the word-wide checker build (`BYTE_CRC=0`, BUILD.WORD_CRC = 1), where every checker takes one beat per cycle as in the RAPIDS Beats build.

Figure 3.1: Byte minus beats utilization, beat-aligned cells



utilization delta

Beat-aligned rows: bytes, beats, efficiency, MB/s and share of the 3200 MB/s peak

Ch	Beats /ch	Sink bytes	Sink beats	Sink eff	Sink MB/s	Sink % peak	Src bytes	Src beats	Src eff	Src MB/s	Src % peak
1	1	32	1	1.000	1600	50.0 %	32	1	1.000	356	11.1 %
1	4	128	4	1.000	2560	80.0 %	128	4	1.000	356	11.1 %
1	16	512	16	1.000	497	15.5 %	512	16	1.000	356	11.1 %
1	64	2048	64	1.000	383	12.0 %	1056	33	1.000	350	10.9 %
1	256	8192	256	1.000	362	11.3 %	4864	152	1.000	355	11.1 %
1	1024	3276	1024	1.00	357	11.2	2944	920	1.00	355	11.1

Ch	Beats /ch	Sink bytes	Sink beats	Sink eff	Sink MB/s	Sink % peak	Src bytes	Src beats	Src eff	Src MB/s	Src % peak
		8		0		%	0		0		%
1	4096	131072	4096	1.000	356	11.1 %	127744	3992	1.000	356	11.1 %
2	1	64	2	1.000	1600	50.0 %	64	2	1.000	356	11.1 %
2	4	256	8	1.000	337	10.5 %	256	8	1.000	356	11.1 %
2	16	1024	32	1.000	415	13.0 %	928	29	1.000	349	10.9 %
2	64	4096	128	1.000	369	11.5 %	1312	41	1.000	351	11.0 %
2	256	16384	512	1.000	359	11.2 %	9152	286	1.000	355	11.1 %
2	1024	65536	2048	1.000	356	11.1 %	58304	1822	1.000	355	11.1 %
2	4096	262144	8192	1.000	356	11.1 %	254912	7966	1.000	356	11.1 %
4	1	128	4	1.000	1600	50.0 %	128	4	1.000	356	11.1 %
4	4	512	16	1.000	413	12.9 %	512	16	1.000	356	11.1 %
4	16	2048	64	1.000	383	12.0 %	1056	33	1.000	352	11.0 %
4	64	8192	256	1.000	360	11.3 %	1792	56	1.000	351	11.0 %
4	256	32768	1024	1.000	357	11.2 %	17472	546	1.000	355	11.1 %
4	1024	131072	4096	1.000	356	11.1 %	115776	3618	1.000	356	11.1 %
4	4096	524288	16384	1.000	356	11.1 %	508992	15906	1.000	356	11.1 %

Ch	Beats /ch	Sink bytes	Sink beats	Sink eff	Sink MB/s	Sink % peak	Src bytes	Src beats	Src eff	Src MB/s	Src % peak
8	1	256	8	1.000	130	4.1 %	256	8	1.000	356	11.1 %
8	4	1024	32	1.000	415	13.0 %	960	30	1.000	347	10.8 %
8	16	4096	128	1.000	369	11.5 %	1280	40	1.000	352	11.0 %
8	64	16384	512	1.000	358	11.2 %	2720	85	1.000	355	11.1 %
8	256	65536	2048	1.000	356	11.1 %	33856	1058	1.000	355	11.1 %
8	1024	262144	8192	1.000	356	11.1 %	230464	7202	1.000	356	11.1 %
8	4096	1048576	32768	1.000	356	11.1 %	1016896	31778	1.000	356	11.1 %

Every beat-aligned row moves whole beats, so efficiency is 1.000 by construction; measured range over 56 rows: 1.000 to 1.000.

4. Transfer size (offset 0)

One descriptor per channel, payload in bytes, start offset 0. Bytes moved and beats are totals over the active channels; efficiency is payload bytes over beats times 32 bytes per beat, on the AXIS side and on the memory side. MB/s is total bytes over the measurement window (the longer of the stream and memory windows) and is always shown with its share of the 3200 MB/s peak (100 MHz x 32 B).

4.1 Sink path

Size sweep, sink path, offset 0

Payload B	Offset	Ch	Bytes moved	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
1	0	1	1	1	1	0.031	0.031	50.0	1.56 %	PASS
1	0	2	2	2	2	0.031	0.031	50.0	1.56 %	PASS
1	0	4	4	4	4	0.031	0.031	50.0	1.56 %	PASS
1	0	8	8	8	8	0.031	0.031	4.1	0.13 %	PASS
7	0	1	7	1	1	0.219	0.219	350.0	10.94 %	PASS
7	0	2	14	2	2	0.219	0.219	350.0	10.94 %	PASS
7	0	4	28	4	4	0.219	0.219	350.0	10.94 %	PASS
7	0	8	56	8	8	0.219	0.219	28.4	0.89 %	PASS
32	0	1	32	1	1	1.000	1.000	1600.0	50.00 %	PASS
32	0	2	64	2	2	1.000	1.000	1600.0	50.00 %	PASS
32	0	4	128	4	4	1.000	1.000	1600.0	50.00 %	PASS
32	0	8	256	8	8	1.000	1.000	129.9	4.06	PASS

Payload B	Offset	Ch	Bytes moved	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
									%	
33	0	1	33	2	2	0.516	0.516	1100.0	34.38 %	PASS
33	0	2	66	4	4	0.516	0.516	1320.0	41.25 %	PASS
33	0	4	132	8	8	0.516	0.516	113.8	3.56 %	PASS
33	0	8	264	16	16	0.516	0.516	129.4	4.04 %	PASS
64	0	1	64	2	2	1.000	1.000	2133.3	66.67 %	PASS
64	0	2	128	4	4	1.000	1.000	2560.0	80.00 %	PASS
64	0	4	256	8	8	1.000	1.000	220.7	6.90 %	PASS
64	0	8	512	16	16	1.000	1.000	251.0	7.84 %	PASS
77	0	1	77	3	3	0.802	0.802	1925.0	60.16 %	PASS
77	0	2	154	6	6	0.802	0.802	208.1	6.50 %	PASS
77	0	4	308	12	12	0.802	0.802	256.7	8.02 %	PASS
77	0	8	616	24	24	0.802	0.802	290.6	9.08 %	PASS
203	0	1	203	7	7	0.906	0.906	369.1	11.53 %	PASS
203	0	2	406	14	14	0.906	0.906	495.1	15.47 %	PASS
203	0	4	812	28	28	0.906	0.906	414.3	12.95 %	PASS

Payload B	Offset	Ch	Bytes moved	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
203	0	8	1624	56	56	0.906	0.906	379.4	11.86 %	PASS
1024	0	1	1024	32	32	1.000	1.000	414.6	12.96 %	PASS
1024	0	2	2048	64	64	1.000	1.000	382.8	11.96 %	PASS
1024	0	4	4096	128	128	1.000	1.000	368.7	11.52 %	PASS
1024	0	8	8192	256	256	1.000	1.000	362.0	11.31 %	PASS
4035	0	1	4035	127	127	0.993	0.993	366.2	11.44 %	PASS
4035	0	2	8070	254	254	0.993	0.993	360.6	11.27 %	PASS
4035	0	4	16140	508	508	0.993	0.993	357.3	11.17 %	PASS
4035	0	8	32280	1016	1016	0.993	0.993	356.3	11.13 %	PASS
4096	0	1	4096	128	128	1.000	1.000	368.7	11.52 %	PASS
4096	0	2	8192	256	256	1.000	1.000	362.0	11.31 %	PASS
4096	0	4	16384	512	512	1.000	1.000	358.7	11.21 %	PASS
4096	0	8	32768	1024	1024	1.000	1.000	357.1	11.16 %	PASS

4.2 Source path

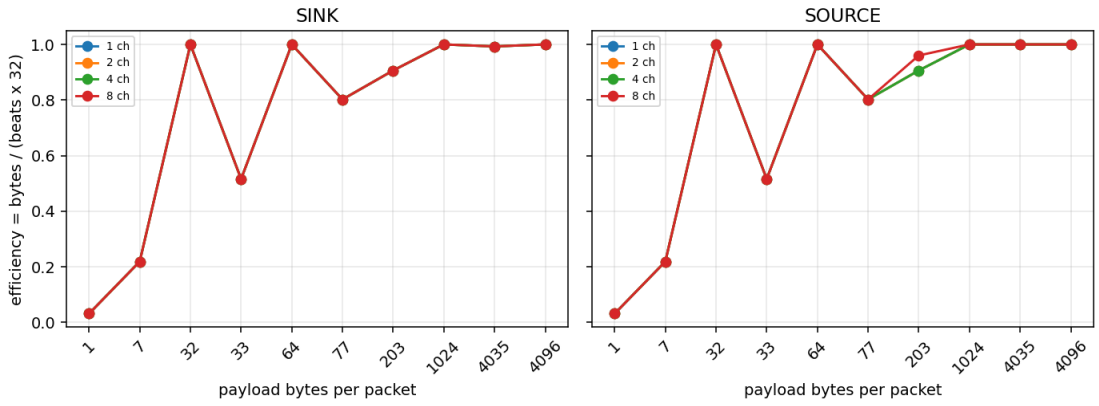
Size sweep, source path, offset 0

Payload B	Offset	Ch	Bytes moved	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
1	0	1	1	1	1	0.031	0.031	11.1	0.35 %	PASS
1	0	2	2	2	2	0.031	0.031	18.2	0.57 %	PASS
1	0	4	4	4	4	0.031	0.031	26.7	0.83 %	PASS
1	0	8	8	8	8	0.031	0.031	34.8	1.09 %	PASS
7	0	1	7	1	1	0.219	0.219	77.8	2.43 %	PASS
7	0	2	14	2	2	0.219	0.219	116.7	3.65 %	PASS
7	0	4	28	4	4	0.219	0.219	155.6	4.86 %	PASS
7	0	8	56	8	8	0.219	0.219	186.7	5.83 %	PASS
32	0	1	32	1	1	1.000	1.000	355.6	11.11 %	PASS
32	0	2	64	2	2	1.000	1.000	355.6	11.11 %	PASS
32	0	4	128	4	4	1.000	1.000	355.6	11.11 %	PASS
32	0	8	256	8	8	1.000	1.000	355.6	11.11 %	PASS
33	0	1	33	2	2	0.516	0.516	183.3	5.73 %	PASS
33	0	2	66	4	4	0.516	0.516	227.6	7.11 %	PASS
33	0	4	132	8	8	0.516	0.516	258.8	8.09	PASS

Payload B	Offset	Ch	Bytes moved	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
33	0	8	264	16	16	0.516	0.516	277.9	8.68 %	PASS
64	0	1	64	2	2	1.000	1.000	355.6	11.11 %	PASS
64	0	2	128	4	4	1.000	1.000	355.6	11.11 %	PASS
64	0	4	256	8	8	1.000	1.000	355.6	11.11 %	PASS
64	0	8	512	16	16	1.000	1.000	355.6	11.11 %	PASS
77	0	1	77	3	3	0.802	0.802	285.2	8.91 %	PASS
77	0	2	154	6	6	0.802	0.802	308.0	9.62 %	PASS
77	0	4	308	12	12	0.802	0.802	320.8	10.03 %	PASS
77	0	8	616	24	24	0.802	0.802	327.7	10.24 %	PASS
203	0	1	203	7	7	0.906	0.906	322.2	10.07 %	PASS
203	0	2	406	14	14	0.906	0.906	335.5	10.49 %	PASS
203	0	4	812	28	28	0.906	0.906	342.6	10.71 %	PASS
203	0	8	1014	33	56	0.960	0.566	347.3	10.85 %	PASS
1024	0	1	928	29	32	1.000	0.906	348.9	10.90 %	PASS
1024	0	2	1056	33	64	1.000	0.516	349.7	10.93 %	PASS

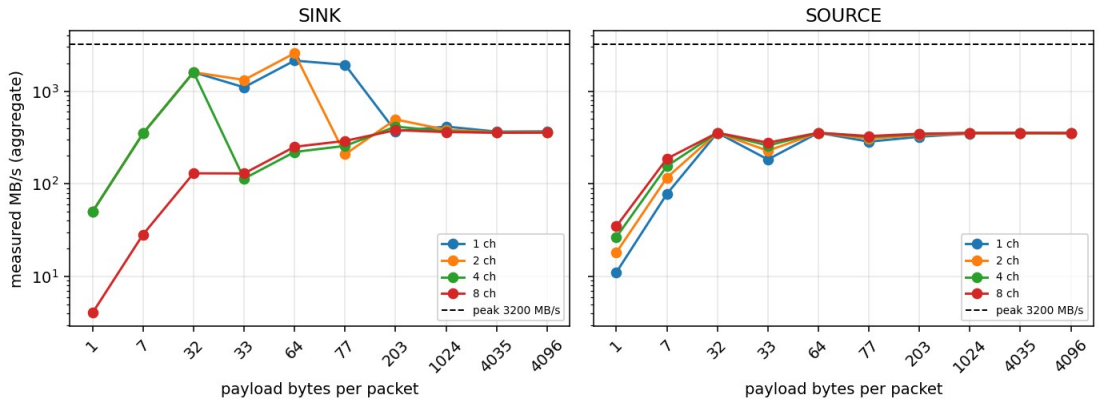
Payload B	Offset	Ch	Bytes moved	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
1024	0	4	1312	41	128	1.000	0.320	354.6	11.08 %	PASS
1024	0	8	1760	55	256	1.000	0.215	353.4	11.04 %	PASS
4035	0	1	1280	40	127	1.000	0.315	350.7	10.96 %	PASS
4035	0	2	1760	55	254	1.000	0.217	352.0	11.00 %	PASS
4035	0	4	2688	84	508	1.000	0.165	352.8	11.02 %	PASS
4035	0	8	4512	141	1016	1.000	0.139	355.3	11.10 %	PASS
4096	0	1	1280	40	128	1.000	0.312	350.7	10.96 %	PASS
4096	0	2	1760	55	256	1.000	0.215	352.0	11.00 %	PASS
4096	0	4	2688	84	512	1.000	0.164	352.3	11.01 %	PASS
4096	0	8	4512	141	1024	1.000	0.138	353.9	11.06 %	PASS

Figure 4.1: Efficiency against payload size



efficiency vs size

Figure 4.2: Measured MB/s against payload size, peak as the dashed line



MB/s vs size

Observations from this data:

- Sink, 8 channels, 4096 B: 357 MB/s = 11.2 % of the 3200 MB/s peak at efficiency 1.000.
- Sink, 8 channels: 1 B payload reaches 4.1 MB/s, 88x less than 4096 B; the window is dominated by fixed per-packet latency (197 cycles for one beat per channel).
- Source, 8 channels, 4096 B: 354 MB/s = 11.1 % of the 3200 MB/s peak at efficiency 1.000.
- Source, 8 channels: 1 B payload reaches 34.8 MB/s, 10x less than 4096 B; the window is dominated by fixed per-packet latency (23 cycles for one beat per channel).

5. Start offset

A descriptor may start mid-beat. The offset moves payload into an extra memory beat when `offset + payload` crosses a beat boundary, lowering memory-side efficiency while the AXIS side, which packs from lane 0, is unchanged. Offset 0 rows repeat the size sweep.

5.1 Sink path

Offset sweep, sink path

Payload B	Offset	Ch	Bytes moved	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
33	0	1	33	2	2	0.516	0.516	1100.0	34.38 %	PASS
33	1	1	33	2	2	0.516	0.516	1100.0	34.38 %	PASS
33	5	1	33	2	2	0.516	0.516	1100.0	34.38 %	PASS
33	31	1	33	2	2	0.516	0.516	1100.0	34.38 %	PASS
203	0	1	203	7	7	0.906	0.906	369.1	11.53 %	PASS
203	1	1	203	7	7	0.906	0.906	369.1	11.53 %	PASS
203	5	1	203	7	7	0.906	0.906	369.1	11.53 %	PASS
203	31	1	203	7	8	0.906	0.793	225.6	7.05 %	PASS
1024	0	1	1024	32	32	1.000	1.000	414.6	12.96 %	PASS
1024	1	1	1024	32	33	1.000	0.970	317.0	9.91 %	PASS
1024	5	1	1024	32	33	1.000	0.970	318.0	9.94 %	PASS
1024	31	1	1024	32	33	1.000	0.970	324.1	10.13 %	PASS
403	0	1	4035	127	127	0.993	0.993	366.2	11.44	PASS

Payload B	Offset	Ch	Bytes moved	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
5									%	
4035	1	1	4035	127	127	0.993	0.993	366.2	11.44 %	PASS
4035	5	1	4035	127	127	0.993	0.993	366.5	11.45 %	PASS
4035	31	1	4035	127	128	0.993	0.985	344.6	10.77 %	PASS
33	0	8	264	16	16	0.516	0.516	129.4	4.04 %	PASS
33	1	8	264	16	16	0.516	0.516	129.4	4.04 %	PASS
33	5	8	264	16	16	0.516	0.516	129.4	4.04 %	PASS
33	31	8	264	16	16	0.516	0.516	129.4	4.04 %	PASS
203	0	8	1624	56	56	0.906	0.906	379.4	11.86 %	PASS
203	1	8	1624	56	56	0.906	0.906	379.4	11.86 %	PASS
203	5	8	1624	56	56	0.906	0.906	380.3	11.89 %	PASS
203	31	8	1624	56	64	0.906	0.793	249.8	7.81 %	PASS
1024	0	8	8192	256	256	1.000	1.000	362.0	11.31 %	PASS
1024	1	8	8192	256	264	1.000	0.970	328.6	10.27 %	PASS
1024	5	8	8192	256	264	1.000	0.970	328.7	10.27 %	PASS
1024	31	8	8192	256	264	1.000	0.970	329.5	10.30 %	PASS

Payload B	Offset	Ch	Bytes moved	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
4035	0	8	32280	1016	1016	0.993	0.993	356.3	11.13 %	PASS
4035	1	8	32280	1016	1016	0.993	0.993	356.3	11.13 %	PASS
4035	5	8	32280	1016	1016	0.993	0.993	356.3	11.14 %	PASS
4035	31	8	32280	1016	1024	0.993	0.985	347.4	10.86 %	PASS

5.2 Source path

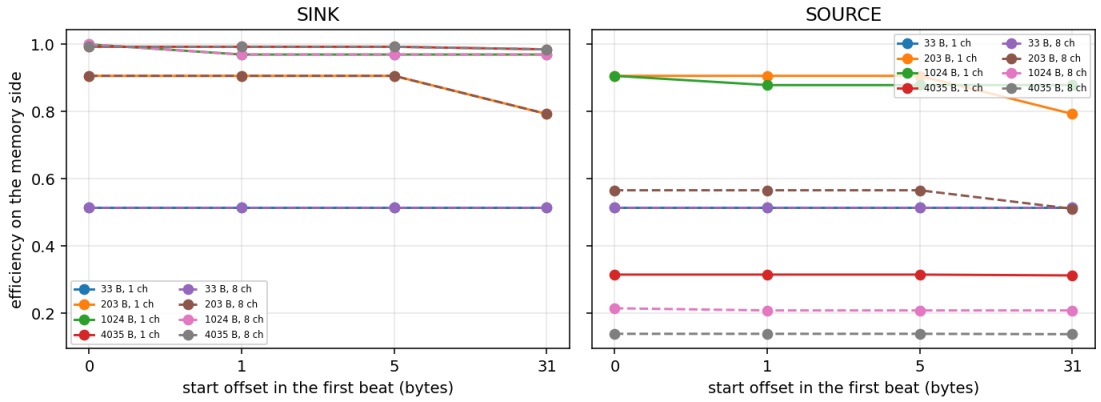
Offset sweep, source path

Payload B	Offset	Ch	Bytes moved	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
33	0	1	33	2	2	0.516	0.516	183.3	5.73 %	PASS
33	1	1	33	2	2	0.516	0.516	173.7	5.43 %	PASS
33	5	1	33	2	2	0.516	0.516	173.7	5.43 %	PASS
33	31	1	33	2	2	0.516	0.516	173.7	5.43 %	PASS
203	0	1	203	7	7	0.906	0.906	322.2	10.07 %	PASS
203	1	1	203	7	7	0.906	0.906	317.2	9.91 %	PASS
203	5	1	203	7	7	0.906	0.906	317.2	9.91 %	PASS
203	31	1	203	7	8	0.906	0.793	317.2	9.91 %	PASS
1024	0	1	928	29	32	1.000	0.906	348.9	10.90 %	PASS

Payload B	Offset	Ch	Bytes moved	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
1024	1	1	928	29	33	1.000	0.879	348.9	10.90%	PASS
1024	5	1	928	29	33	1.000	0.879	348.9	10.90%	PASS
1024	31	1	928	29	33	1.000	0.879	348.9	10.90%	PASS
4035	0	1	1280	40	127	1.000	0.315	350.7	10.96%	PASS
4035	1	1	1280	40	127	1.000	0.315	350.7	10.96%	PASS
4035	5	1	1280	40	127	1.000	0.315	350.7	10.96%	PASS
4035	31	1	1280	40	128	1.000	0.312	350.7	10.96%	PASS
33	0	8	264	16	16	0.516	0.516	277.9	8.68%	PASS
33	1	8	264	16	16	0.516	0.516	275.0	8.59%	PASS
33	5	8	264	16	16	0.516	0.516	275.0	8.59%	PASS
33	31	8	264	16	16	0.516	0.516	275.0	8.59%	PASS
203	0	8	1014	33	56	0.960	0.566	347.3	10.85%	PASS
203	1	8	1014	33	56	0.960	0.566	347.3	10.85%	PASS
203	5	8	1014	33	56	0.960	0.566	347.3	10.85%	PASS
203	31	8	1046	34	64	0.961	0.511	352.2	11.01%	PASS
1024	0	8	1760	55	256	1.000	0.215	353.4	11.04%	PASS

Payload B	Offset	Ch	Bytes moved	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
1024	1	8	1760	55	264	1.000	0.208	349.9	10.93 %	PASS
1024	5	8	1760	55	264	1.000	0.208	349.9	10.93 %	PASS
1024	31	8	1760	55	264	1.000	0.208	349.9	10.93 %	PASS
4035	0	8	4512	141	1016	1.000	0.139	355.3	11.10 %	PASS
4035	1	8	4512	141	1016	1.000	0.139	355.3	11.10 %	PASS
4035	5	8	4512	141	1016	1.000	0.139	355.3	11.10 %	PASS
4035	31	8	4512	141	1024	1.000	0.138	353.9	11.06 %	PASS

Figure 5.1: Memory-side efficiency against start offset



efficiency vs offset

Offsets that cost an extra memory beat (sink, 1 channel): 203 B at offset 31: 7 -> 8 memory beats; 1024 B at offset 1: 32 -> 33 memory beats; 1024 B at offset 5: 32 -> 33 memory beats; 1024 B at offset 31: 32 -> 33 memory beats; 4035 B at offset 31: 127 -> 128 memory beats.

6. Descriptor chains

Several descriptors per channel back to back (4 channels), offset 0. Efficiency counts payload over all beats of the chain.

Descriptor chains, 4 channels

Payload B	Descs	Path	Bytes moved	AXIS beats	Eff (AXIS)	MB/s	% of peak	Result
1	16	sink	64	64	0.031	5.4	0.17 %	PASS
1	16	source	64	64	0.031	38.8	1.21 %	PASS
64	4	sink	1024	32	1.000	223.6	6.99 %	PASS
64	4	source	1024	32	1.000	355.6	11.11 %	PASS
64	16	sink	4096	128	1.000	221.4	6.92 %	PASS
64	16	source	3872	121	1.000	354.9	11.09 %	PASS
1024	4	sink	16384	512	1.000	342.5	10.70 %	PASS
1024	4	source	2656	83	1.000	352.3	11.01 %	PASS
1024	16	sink	65536	2048	1.000	336.6	10.52 %	PASS
1024	16	source	50368	1574	1.000	355.6	11.11 %	PASS

7. Source backpressure (integrity only)

The source backpressure is paced by the host toggling the checker's ready over UART, so the cycle counts measure the host, not the DUT (rapids TASK-085). These rows are integrity checks: bytes and beats must still match and the golden CRCs must agree. Their MB/s is not reported.

Source with backpressure, integrity rows

Ch	Payload B	Offset	Expected bytes	Expected AXIS beats	MB/s	Result
1	33	0	33	2	n/a (host-paced)	PASS
1	33	5	33	2	n/a (host-paced)	PASS
1	1024	0	1024	32	n/a (host-paced)	PASS
1	1024	5	1024	32	n/a (host-paced)	PASS
4	33	0	132	8	n/a (host-paced)	PASS
4	33	5	132	8	n/a (host-paced)	PASS
4	1024	0	4096	128	n/a (host-paced)	PASS
4	1024	5	4096	128	n/a (host-paced)	PASS

8. Failures, caveats and coverage

No measured point failed.

Coverage: 117 of 117 planned points of profile `standard` are in this file (117 passed, 0 failed).

Standing limitations of the design, not of this measurement:

- The sink `s_axis_tready` is one signal qualified by TID, so a beat for a channel whose packet record has not arrived blocks every channel behind it on the stream (head-of-line blocking, inherent and documented).
- TYPE=EXT descriptors stay beat-aligned by design and are not part of the byte sweeps.
- AXI error responses are now exercised and the paths are proven on silicon (rapids TASK-020, 2026-10-01); this entry previously recorded them as unproven and the hook as unbuilt. See section 8.1.
- Each interface has its own measurement window, opened on that interface's first handshake; the `sin` window opens on the first ACCEPTED stream beat, and cycles where the stream is offered before that accept are counted by `OBS_SIN_LAUNCH` (section 3.1). MB/s here uses the longer of the stream and memory windows.

8.1 AXI response-error injection (rapids TASK-020)

The two synthetic AXI slaves are built `ERR_INJECT=1` in this harness, so a chosen burst on a chosen channel answers SLVERR instead of OKAY. `ERR_INJ` (0x0C8) arms it – WR_EN for the sink's B channel, RD_EN for the source's R channel, with RESP, CH, SKIP and ONESHOT – and `ERR_STAT` (0x0CC) reports from the SLAVE that the error was really issued, so a passing status check cannot be the DUT agreeing with itself. `ERR_INJECT=0` everywhere else keeps every other consumer answering OKAY as before.

Run	Result
Board, `--byte-seq axi_resp_error --seq-level full`, 8 channels	**36/36 checks PASS**, 7,969 UART ops
UART sim, same sequence at gate	18/18 checks PASS
Bitstream	sha256 `754c3c2d...`, WNS +**0.401 ns** at 100 MHz, 78,426 LUTs (38.5 %), 68 BRAM tiles

Checked per half, per round, with a second channel running alongside throughout: the slave issued the error (`ERR_STAT`); the sticky per-channel flag raised on the TARGETED channel only (`SNK_SCHERR` / `SRC_SCHERR` and `SCHED_ERROR`); `CHANNEL_RESET` cleared it; and both channels were golden against the byte-wise CRC afterwards. So the DUT's sticky `r_wr_error` (BRESP) and `r_rd_error` (RRESP) paths, and recovery through channel reset, are measured on hardware rather than inferred from unit simulation.

NOT covered, and deliberately not claimed: the monbus error PACKET. The host has no monbus-buffer readout – `MON_BASE`/`MON_LIMIT` are configured and never read back – so the packet half of the error contract needs a readout that does not exist yet. rapids TASK-020 keeps that box open.

8.2 Build variants measured against each other (2026-10-01)

One harness RTL; a variant is generic overrides, built by the named targets `make bitstream-{std,perf,mon,obs,ila}`. Every number below is parsed from that build's own post-route reports by `reports/extract_build_metrics.py` – none is typed in – and every variant was then programmed and run on the board.

Variant	LUT	vs std	BRAM	WNS ns	WHS ns	Failing eps	sha256
std (BYTE_CRC=1)	78,426	–	68	0.401	0.045	0	754c3c2dcdb7
perf (BYTE_CRC=0)	71,643	-6,783	52	0.426	0.037	0	1f268d98969d
mon (USE_AXI_MONITORS=1 GEN_MON=1)	84,190	+**5,764**	68	0.768	0.026	0	0678e8d52a88
obs (USE_OBSERVERS=1 + taps)	100,396	+**21,970**	68	0.215	0.031	0	4ff79d23b873

All four close timing with zero failing endpoints out of ~264 k.

What the comparison is for – the cost of instrumentation, which was previously unmeasured on the byte design:

- **The interface observers are expensive**: +21,970 LUTs, +28 % over std, and they more than halve the timing margin (0.401 -> 0.215 ns). That is the one variant where the margin is thin enough to matter on a tighter part.
- **The in-core monitors are cheaper than the observers by 4x** (+5,764 LUTs) and the build closed BETTER than std (0.768 ns). Place-and-route

variation is part of that, so read it as “no margin cost”, not as an improvement.

- **The word-wide perf build is the cheapest in both LUTs and BRAM** (52 vs 68 tiles): the byte-wise CRC machinery is what costs the BRAM, which is the resource price of measuring integrity rather than rate.

Board results, same bitstreams:

Variant	Runs
std / mon / obs	byte smoke PASS, `axi_resp_error` **18/18**, byte-perf quick 7/7
perf	aligned profile **28/28**, 3182 MB/s sink / 3199 MB/s source

The perf row reproduces section 1’s headline figures with a clean identity record on both ends of the run. An earlier attempt produced the same numbers but false-failed its END identity check – the Genesys 2 chain transiently lists the board with no device behind it – which marked good results “not from one board”; `board_guard` now re-reads the chain once before failing, and a test covers both that recovery and that a genuinely absent board still fails.

`perf` deliberately runs only the aligned profile: the host refuses byte-wise checks on a word-wide build (“cannot check a partial strobe”), which is correct and is why the integrity and rate measurements need two bitstreams.

9. Build and resources

Post-route resources and timing at 100 MHz, XC7K325T (203,800 LUTs, 445 BRAM tiles)

Build	Slice LUTs	Slice regs	BRAM tiles	WNS setup	WHS hold	Failing endpoints	Bitstream sha256
std	79,129	57,735	68	+0.726 ns	+0.013 ns	0	`c6985f762eee1e6b`
perf	72,314	54,914	52	+0.258 ns	+0.053 ns	0	`053fc2c192b869c8`
mon	104,032	82,509	68	+0.188 ns	+0.010 ns	0	`fdaeba896348de86`
obs	101,265	90,925	68	+0.044 ns	+0.005 ns	0	`c71a86de44366b38`

Build configuration of each row:

- **std**: BYTE_CRC=1 USE_AXI_MONITORS=0 GEN_MON=0 USE_OBSERVERS=0. RTL commit `f2be469ac`.
- **perf**: BYTE_CRC=0 USE_AXI_MONITORS=0 GEN_MON=0 USE_OBSERVERS=0. RTL commit `616a93077`.
- **mon**: BYTE_CRC=1 USE_AXI_MONITORS=1 GEN_MON=1 USE_OBSERVERS=0 MON_CAPTURE=1. RTL commit `ef18561c9`.
- **obs**: BYTE_CRC=1 USE_OBSERVERS=1 OBS_ENABLE_MON_TAPS=1. RTL commit `ef18561c9`.

Every build closes timing at 100 MHz with no failing endpoint. The slack is small and positive; treat 100 MHz as the design point, not as margin. The resource figures come from the post-route utilization report; the word-wide checker build is a measurement bitstream and is not the standard one.

10. Provenance and reproduction

Provenance

Item	Value
Results file	`rapids_byte_perf_20261003_122406.json`
Timestamp	2026-10-03T12:24:13
Profile	`standard`
Status	final
Bitstream	`rapids_byte.bit` sha256 `053fc2c192b869c8`
Design	256-bit, 8 channels, 4096 B SRAM per channel, 100 MHz, peak 3200 MB/s per direction
Beats reference	`genesys_dw256_obs_C.json`

Device readback per session

Session	Bitstream sha256	CSR_ID	BUILD	Sentinel start	Sentinel end	Stable	Aborted
1	`053fc2c192b869c8`	0x52415042	0x08070820	0x00000000	0x00000000	yes	no

The sentinel is the MON_LIMIT register written once at configure time. It resets on any FPGA reconfiguration, so a change between readbacks means the device was reprogrammed during the run. A point measured across such a change is dropped, not recorded. Device stable for the whole run: yes.

One command runs the campaign and regenerates this report:

```
cd projects/fpga-systems/Genesys2/dma-ip/rapids/flows-rapids
./byte_perf.sh --profile full --final      # final numbers (standard byte-CRC bitstream)
./byte_perf.sh --profile standard          # quick run, writes *_prelim_*.json
```

The word-wide aligned results are `rapids_byte_perf_20261003_121944.json`; its device readback follows.

Device readback per session

Session	Bitstream sha256	CSR_ID	BUILD	Sentinel start	Sentinel end	Stable	Aborted
1	`053fc2c192b869c8`	0x52415042	0x18070820	0x00000000	0x00000000	yes	no

The sentinel is the MON_LIMIT register written once at configure time. It resets on any FPGA reconfiguration, so a change between readbacks means the device was reprogrammed during the run. A point measured across such a change is dropped, not recorded. Device stable for the whole run: yes.